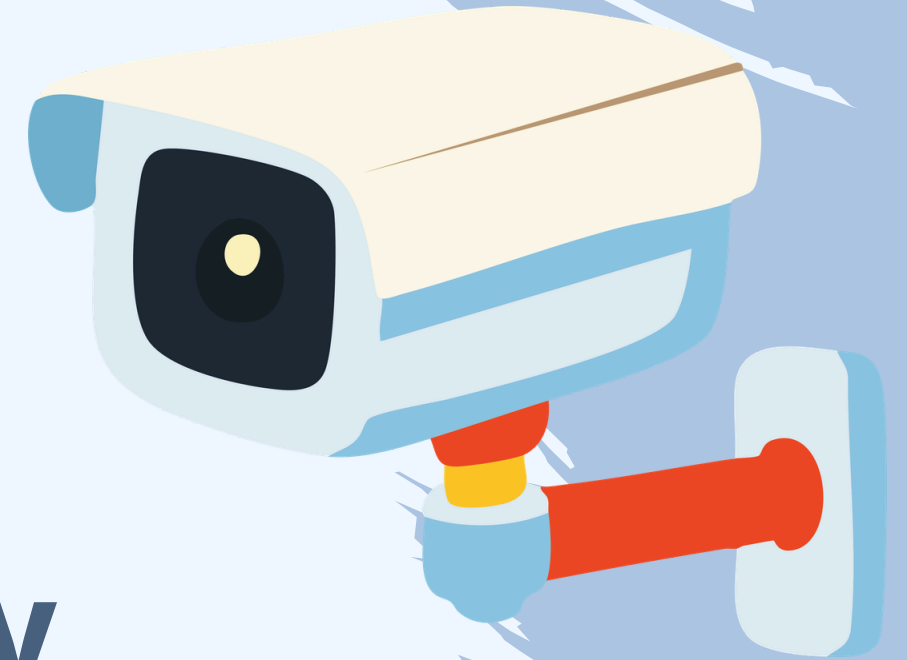


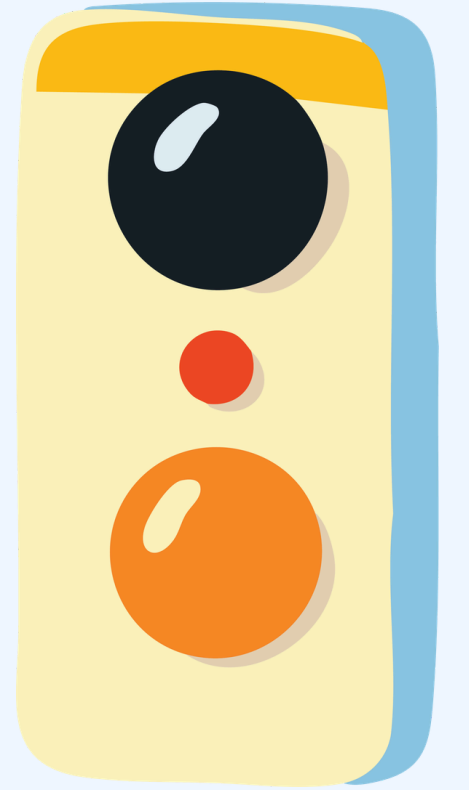
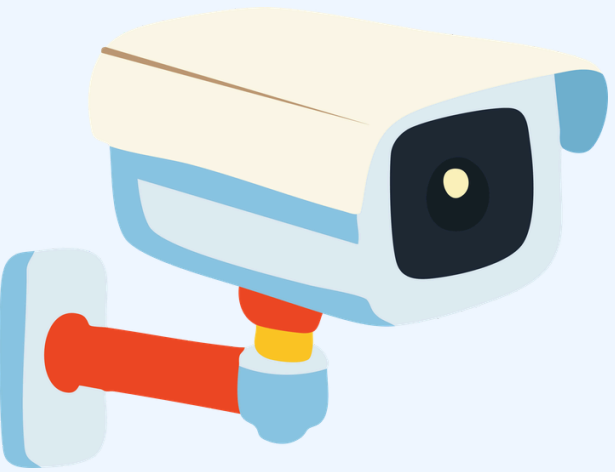
The Effects of Nearby Surveillance on High School Graduation Rates

Noah Carey



Research Question

Are high school graduation rates affected by the prevalence of nearby security cameras?



Methodology



Two Main Data Sets

Baltimore City School Zones
Baltimore County High School Districts
D.C. High School Attendance Zones

Baltimore City School Zones Graduation Rates
Baltimore County High School Graduation Rate
D.C. High School Graduation Rates

&

Baltimore City Cameras
Baltimore County Cameras
D.C. Security Cameras

Data Sources:

- Baltimore City School Zones: Open Baltimore
- Baltimore County School Districts: Baltimore County Government
- D.C. Attendance Zones: Open Data DC
- Baltimore City Graduation Rates: Open Baltimore
- Baltimore County Graduation Rates: Maryland State Department of Education
- D.C. Graduation Rates: D.C. Public Schools
- Baltimore City Cameras: Open Baltimore
- Baltimore County Cameras: Baltimore County Open Data + Directly from County GIS Team
- D.C. Cameras: Surveillance Under Surveillance
- Baltimore City School Locations: Open Baltimore
- Baltimore County School Locations: Baltimore County Open Data
- D.C. School Locations: Open Data DC

Methodology

Step 1: Import
Shapefiles

Step 2: Import
Graduation Rates

Step 3: Import Security
Cameras

Step 4: Import School
Locations

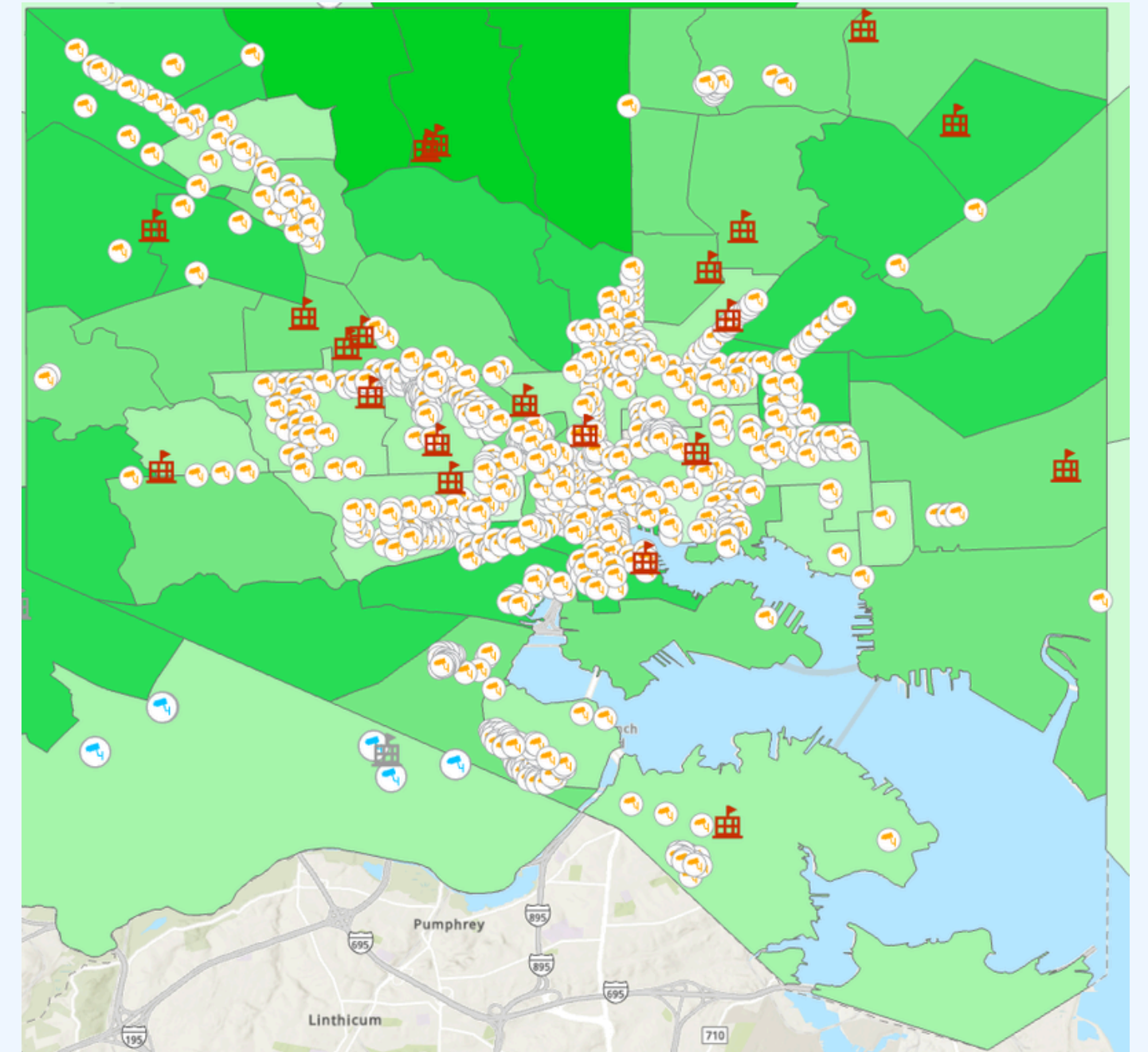
Data Caveats:

All data is for 2023

MacArthur High School was renamed
Duke Ellington High School

Baltimore County's graduation rates
had to be input manually
D.C.'s Cameras had to be
downloaded in three chunks and
converted to points from a CSV and
then combined

The elementary and middle schools
from Baltimore City and D.C. had to
be trimmed



Analysis

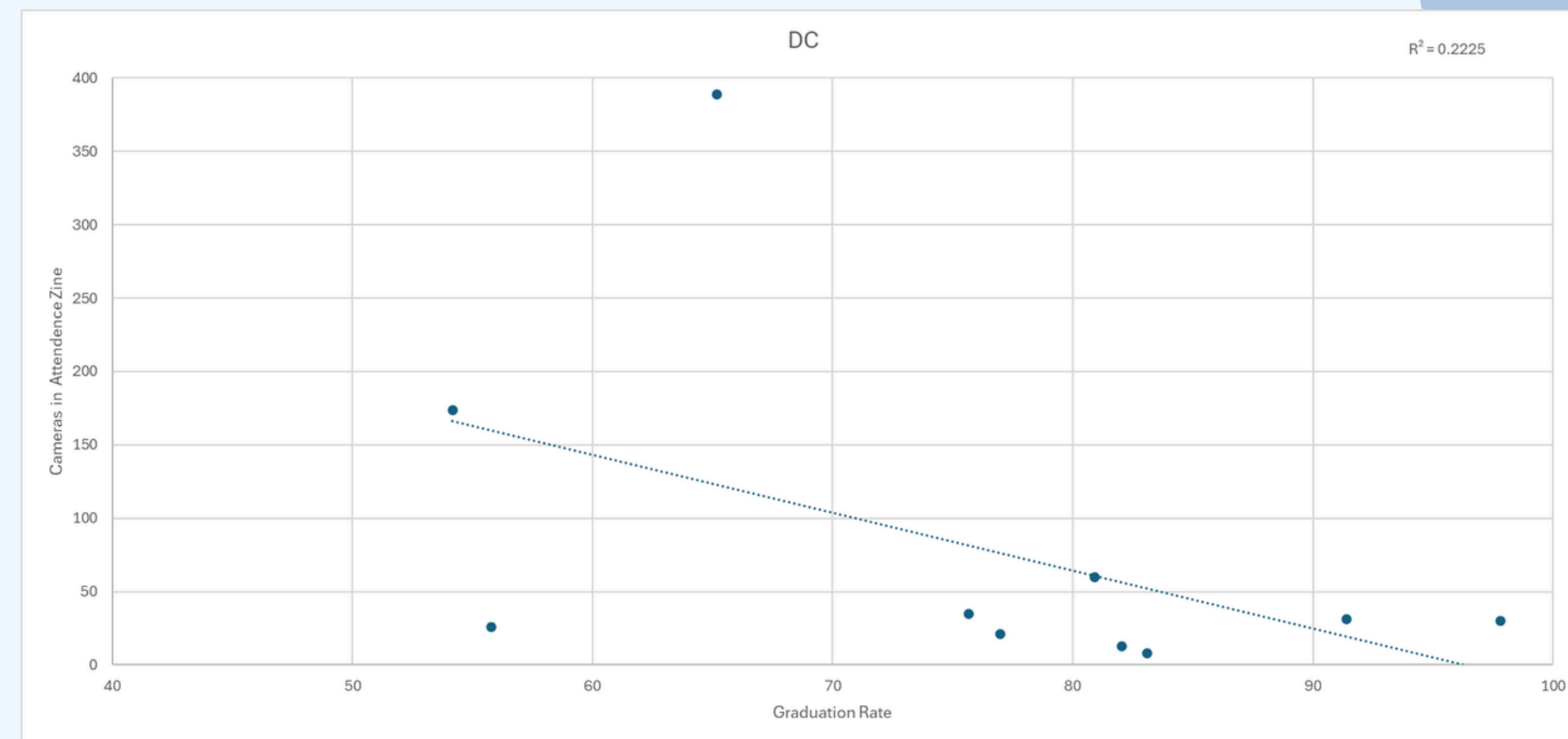
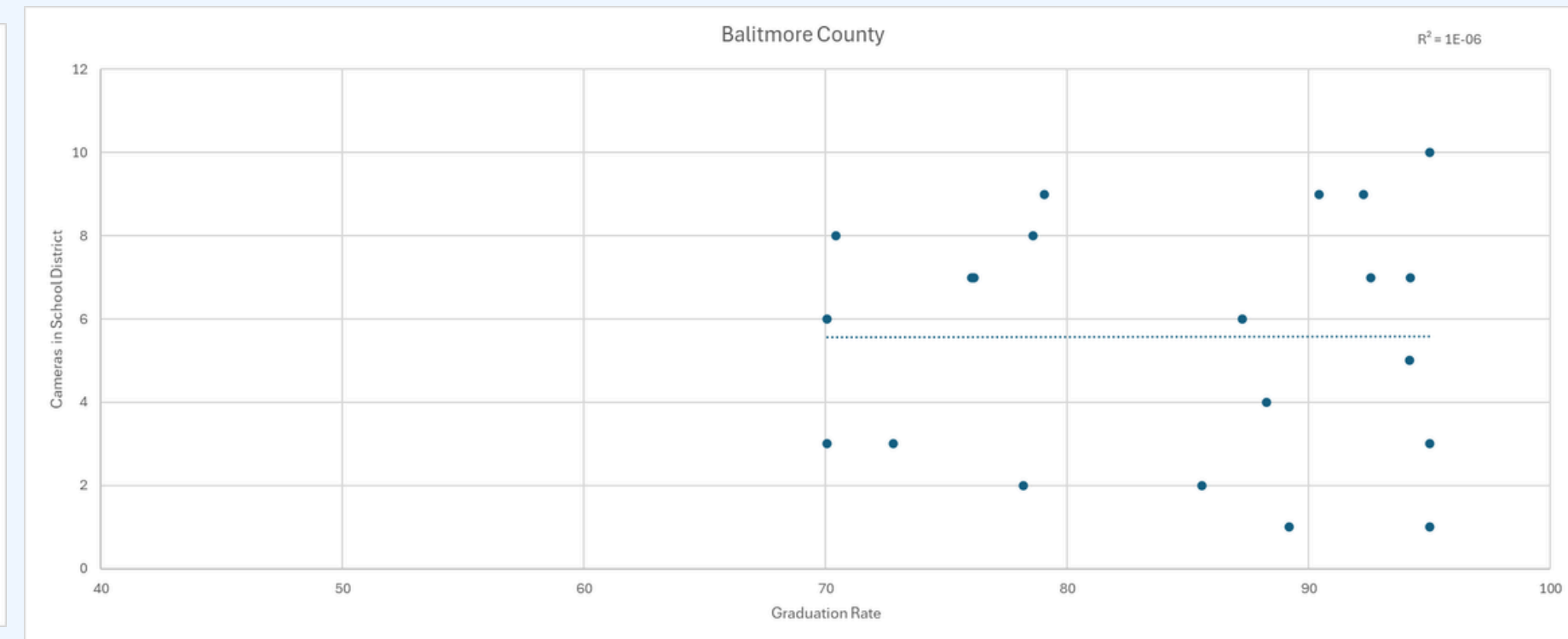
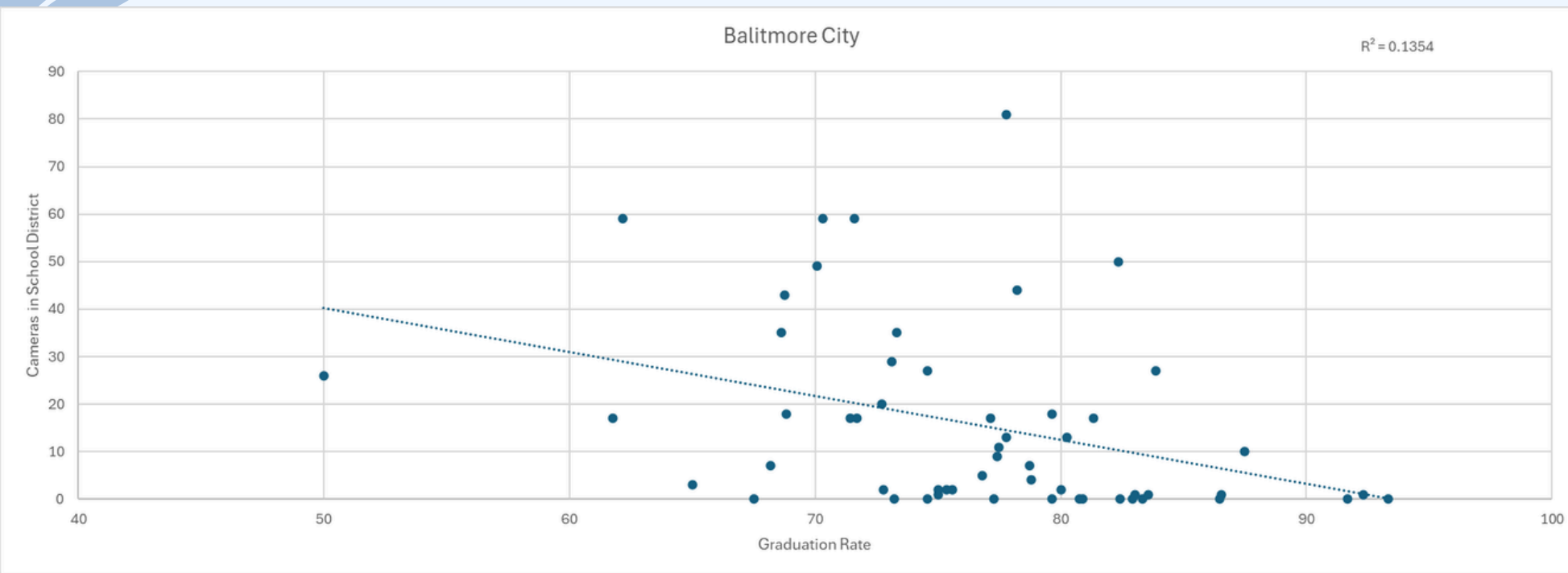
Summarize Within: Used to find out how many cameras were located in each zone

New table was exported as an excel

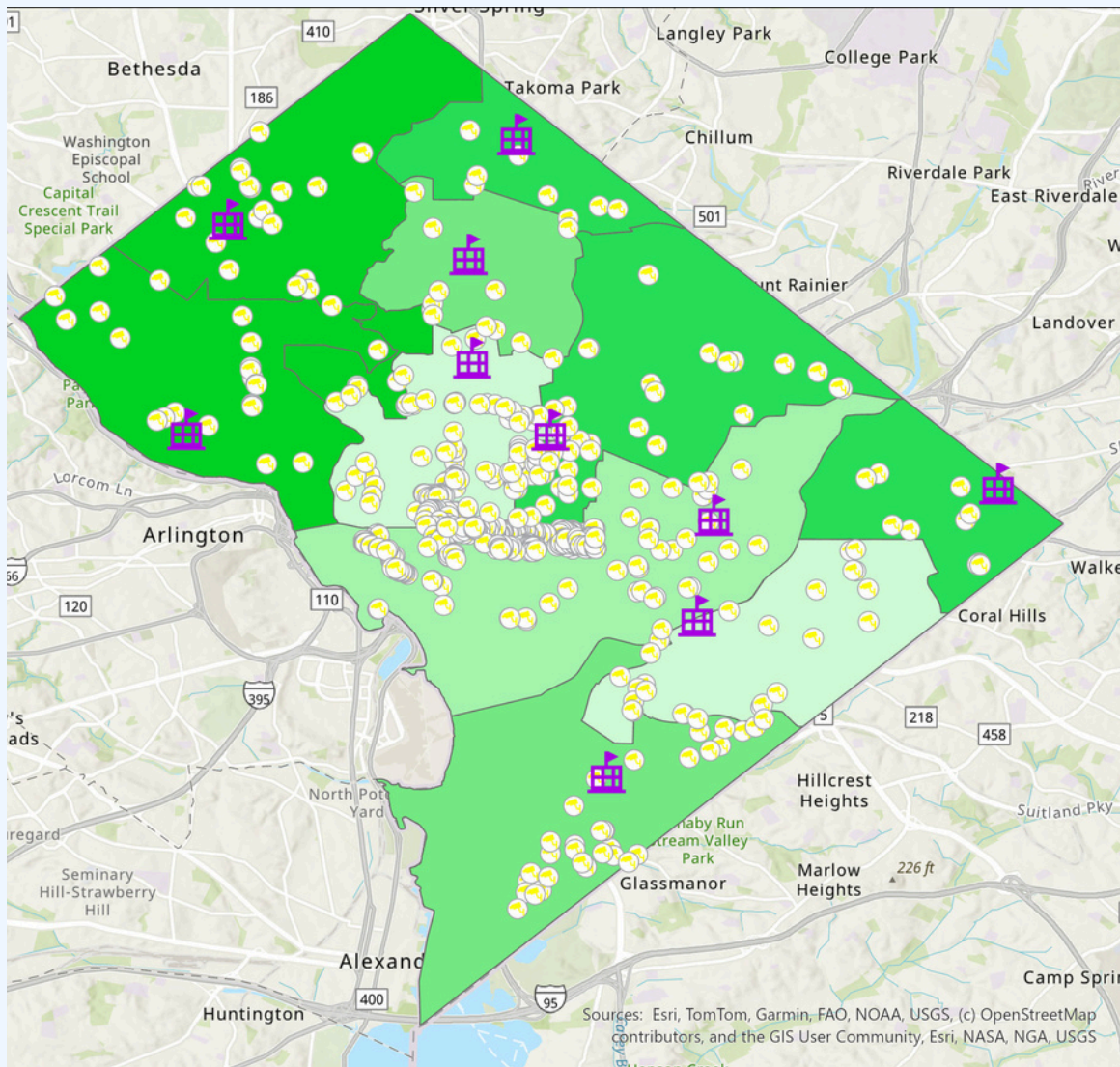
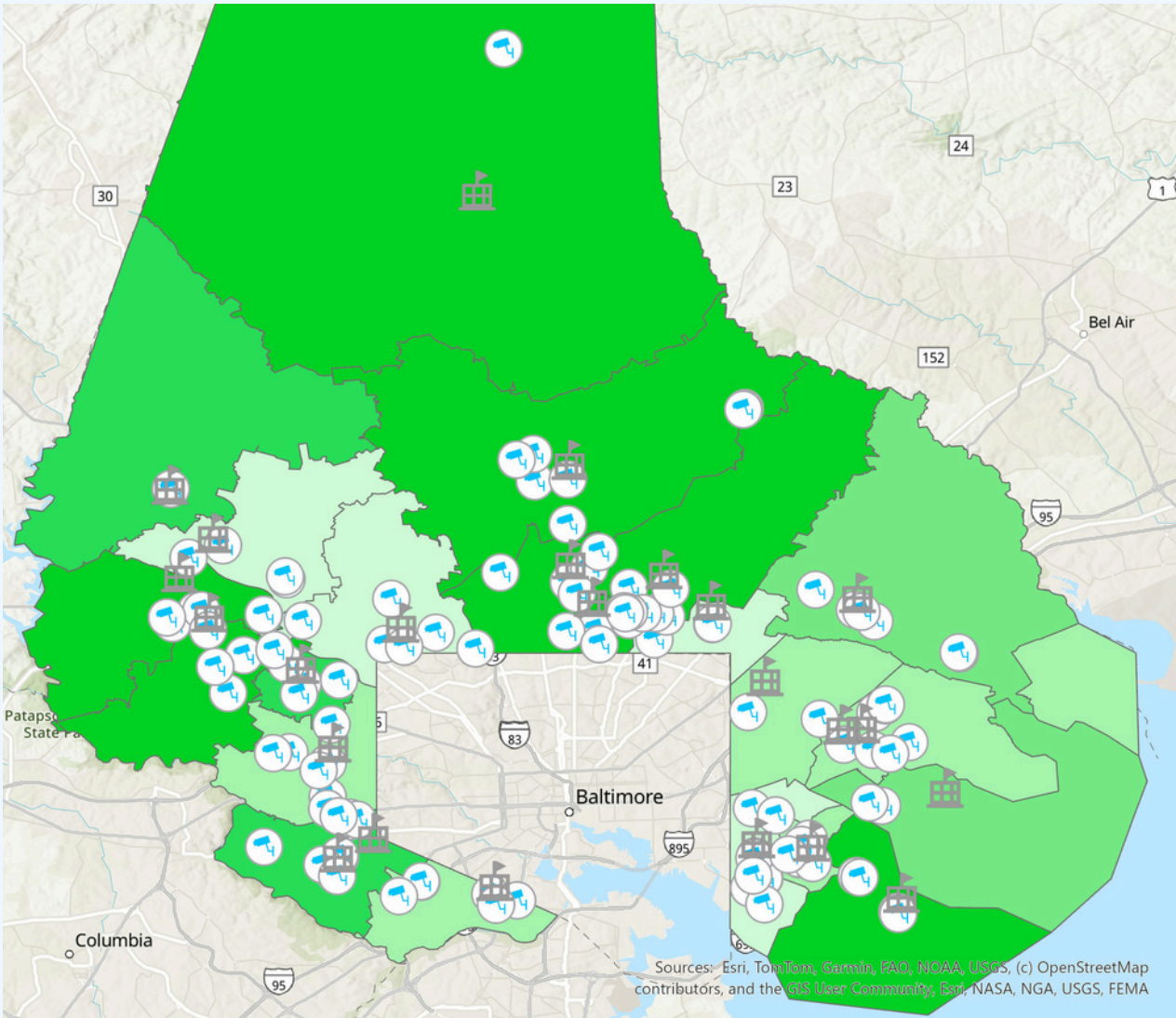
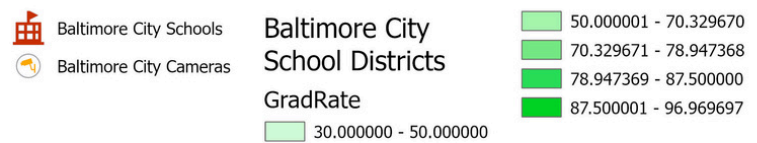
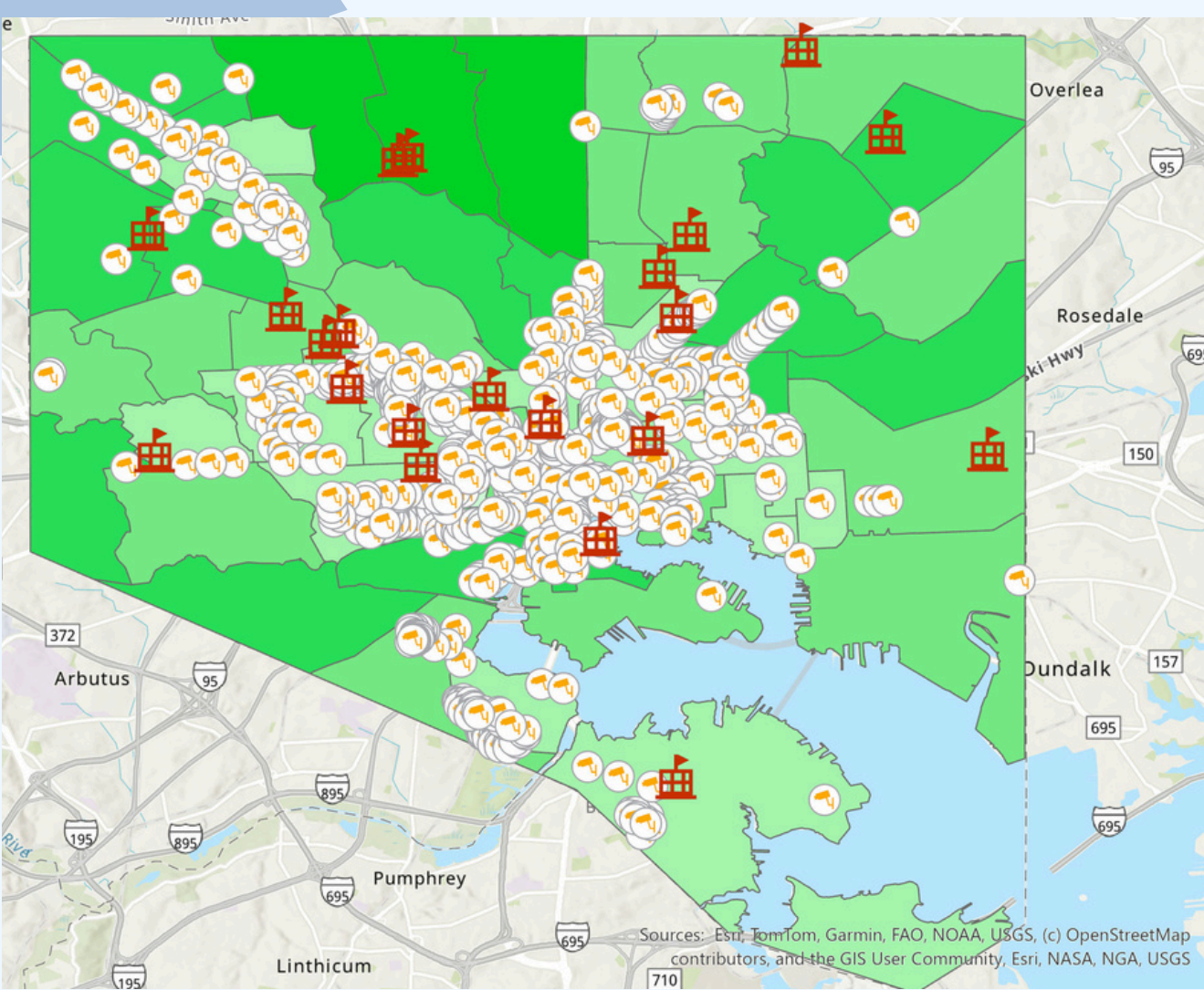
Regression analysis was conducted on the resulting dataset

DC			Baltimore City			Baltimore County		
High School	Graduation Rate	Cameras per District	High School	Graduation Rate	Cameras per District	High School	Graduation Rate	Cameras per District
Jackson-Re	91.40000153	31	Allendale/Irvington/S. Hilton	72.79411765	2	Towson HS	95	10
Ballou	75.68000031	35	Beechfield/Ten Hills/West Hills	83.33333333	0	Perry Hall HS	87.23999786	6
Roosevelt	76.98999786	21	Belair-Edison	80.24691358	13	Kenwood HS	76.12999725	7
Dunbar	80.91999817	60	Brooklyn/Curtis Bay/Hawkins Point	68.83168883	18	Dundalk HS	70.41000366	8
Eastern	65.16999817	389	Canton	65	3	Chesapeake HS	85.58999634	2
Anacostia	55.77000046	26	Cedonia/Frankford	77.27272727	0	Franklin HS	89.18000031	1
Ellington	97.80999756	30	Cherry Hill	62.16216216	59	Hereford HS	95	1
Woodson	82.05000305	13	Chinquapin Park/Belvedere	73.23943662	0	Catonsville HS	90.41999817	9
Cardozo	54.16999817	174	Claremont/Armistead	80.8988764	0	Sparrows Point HS	95	3
Coolidge	83.08000183	8	Clifton-Berea	79.62962963	18	Owings Mills HS	72.80999756	3
			Cross-Country/Cheswolde	87.5	10	Randallstown HS	94.16999817	5
			Dickeyville/Franklintown	80	2	Overlea HS	78.16999817	2
			Dorchester/Ashburton	78.72340426	7	Parkville HS	70.06999969	3
			Downtown/Seton Hill	77.77777778	81	New Town HS	94.22000122	7
			Edmondson Village	75.58139535	2	Woodlawn HS	79.05999756	9
			Fells Point	72.72727273	20	Pikesville HS	70.06999969	6
			Forest Park/Walbrook	80.76923077	0	Milford Mill Acader	88.23999786	4
			Glen-Fallstaff	77.77777778	13	Dulaney HS	92.56999969	7
			Greater Charles Village/Barclay	73.33333333	35	Loch Raven HS	92.26999664	9
			Greater Govans	82.4	0	Lansdowne HS	76.05000305	7
			Greater Mondawmin	71.71717172	17	Patapsco HS	78.56999969	8
			Greater Roland Park/Poplar Hill	91.66666667	0			
			Greater Rosemont	78.21782178	44			
			Greenmount East	81.33333333	17			
			Hamilton	86.55462185	1			
			Harbor East/Little Italy	71.42857143	17			
			Harford/Echodale	79.66101695	0			
			Highlandtown	75	1			
			Howard Park/West Arlington	83.56164384	1			
			Inner Harbor/Federal Hill	82.35294118	50			
			Lauraville	83.03571429	1			
			Loch Raven	77.46478873	11			
			Madison/East End	74.57627119	27			
			Medfield/Hampden/Woodberry/Reming	82.92682927	0			
			Midtown	68.62745098	35			
			Midway/Coldstream	61.76470588	17			
			Morrell Park/Violetville	86.48648649	0			
			Mount Washington/Coldspring	93.33333333	0			
			North Baltimore/Guilford/Homeland	92.30769231	1			
			Northwood	74.56140351	0			
			Oldtown/Middle East	68.75	43			
			Orangeville/East Highlandtown	76.8115942	5			
			Patterson Park North & East	67.5	0			
			Penn North/Reservoir Hill	77.41935484	9			
			Pimlico/Arlington/Hilltop	68.18181818	7			
			Poppleton/The Terraces/Hollins Marke	50	26			
			Sandtown-Winchester/Harlem Park	70.32967033	59			
			South Baltimore	75	2			
			Southeastern	75.34246575	2			
			Southern Park Heights	77.14285714	17			
			Southwest Baltimore	70.08547009	49			
			The Waverlies	78.78787879	4			
			Upton/Druid Heights	71.60493827	59			
			Washington Village/Pigtown	83.87096774	27			
			Westport/Mount Winans/Lakeland	73.13432836	29			

Analysis



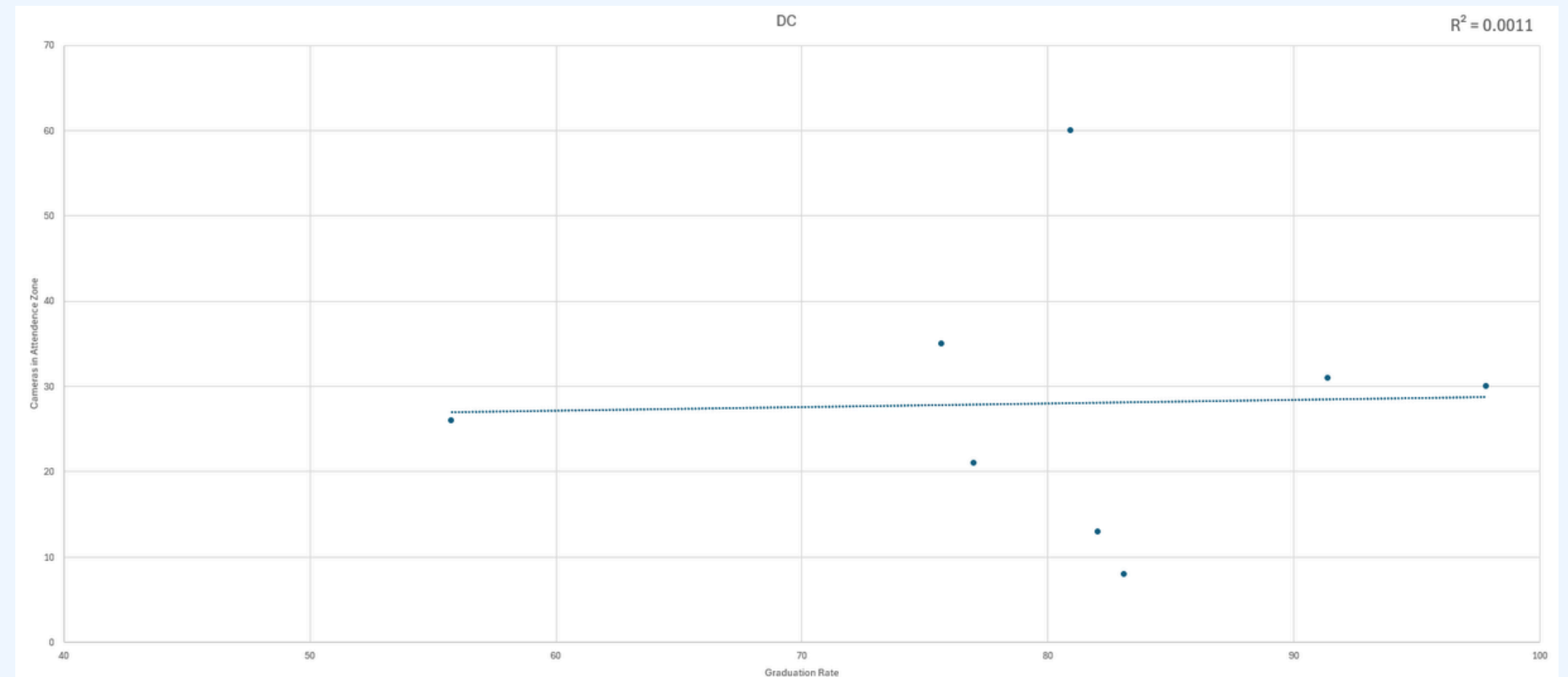
Analysis



Results

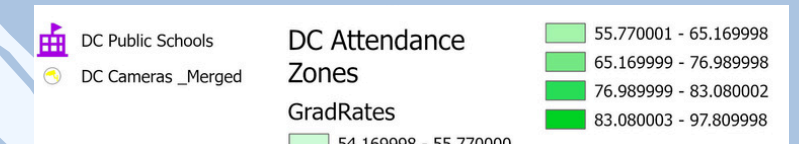
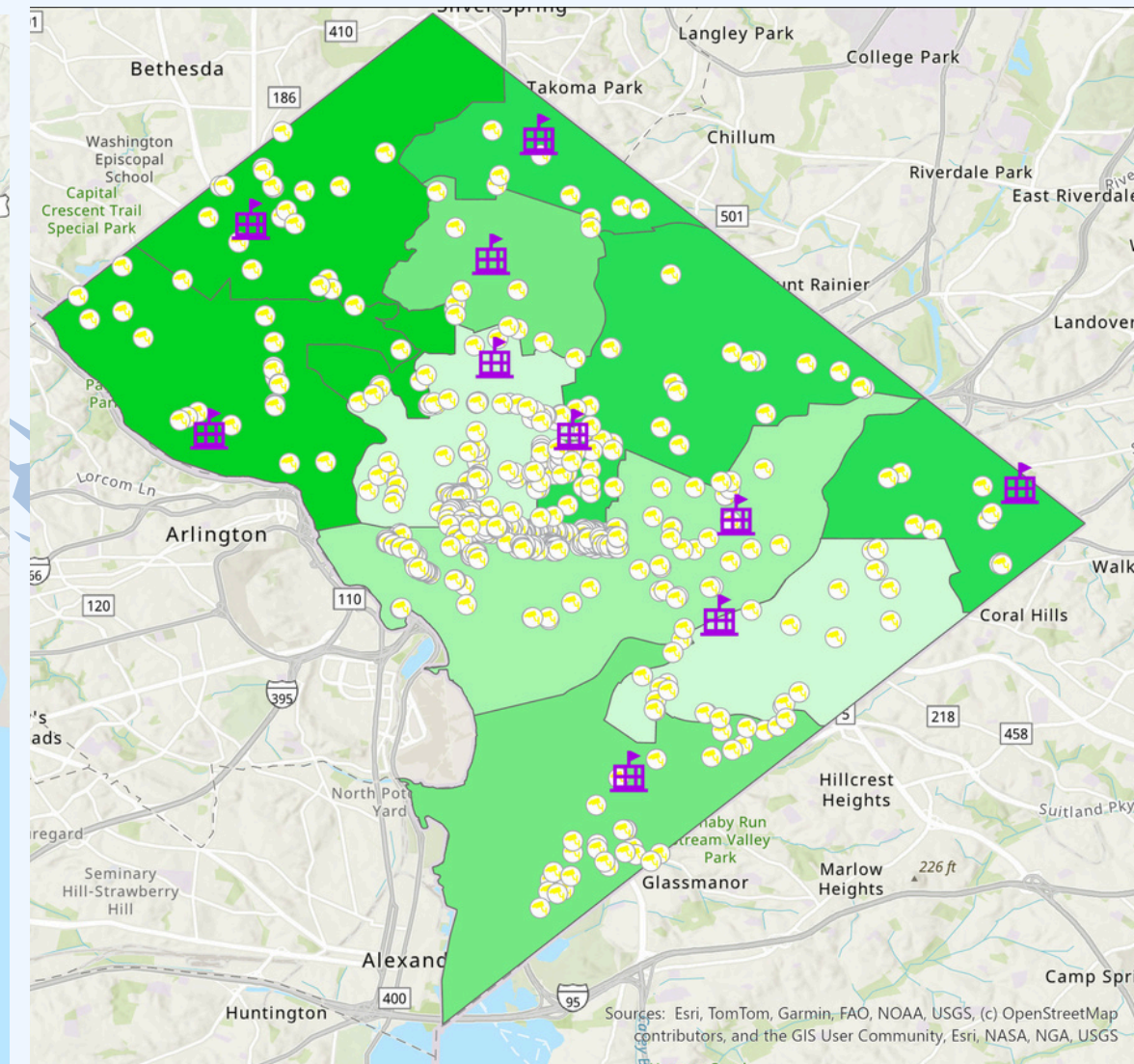
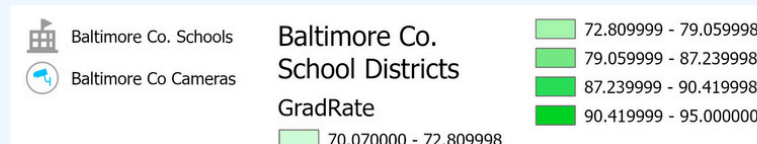
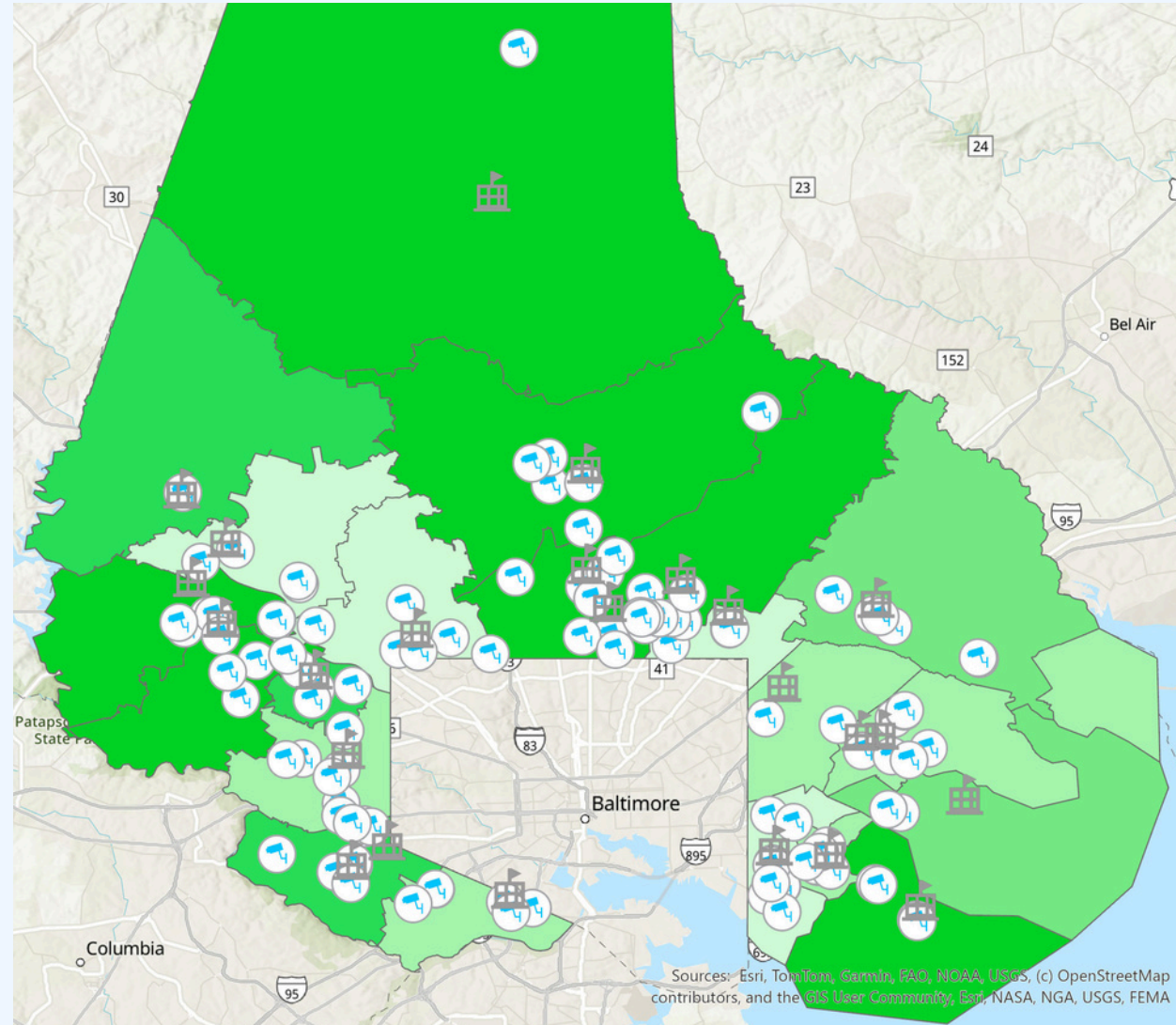
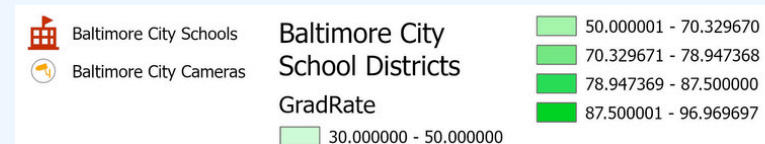
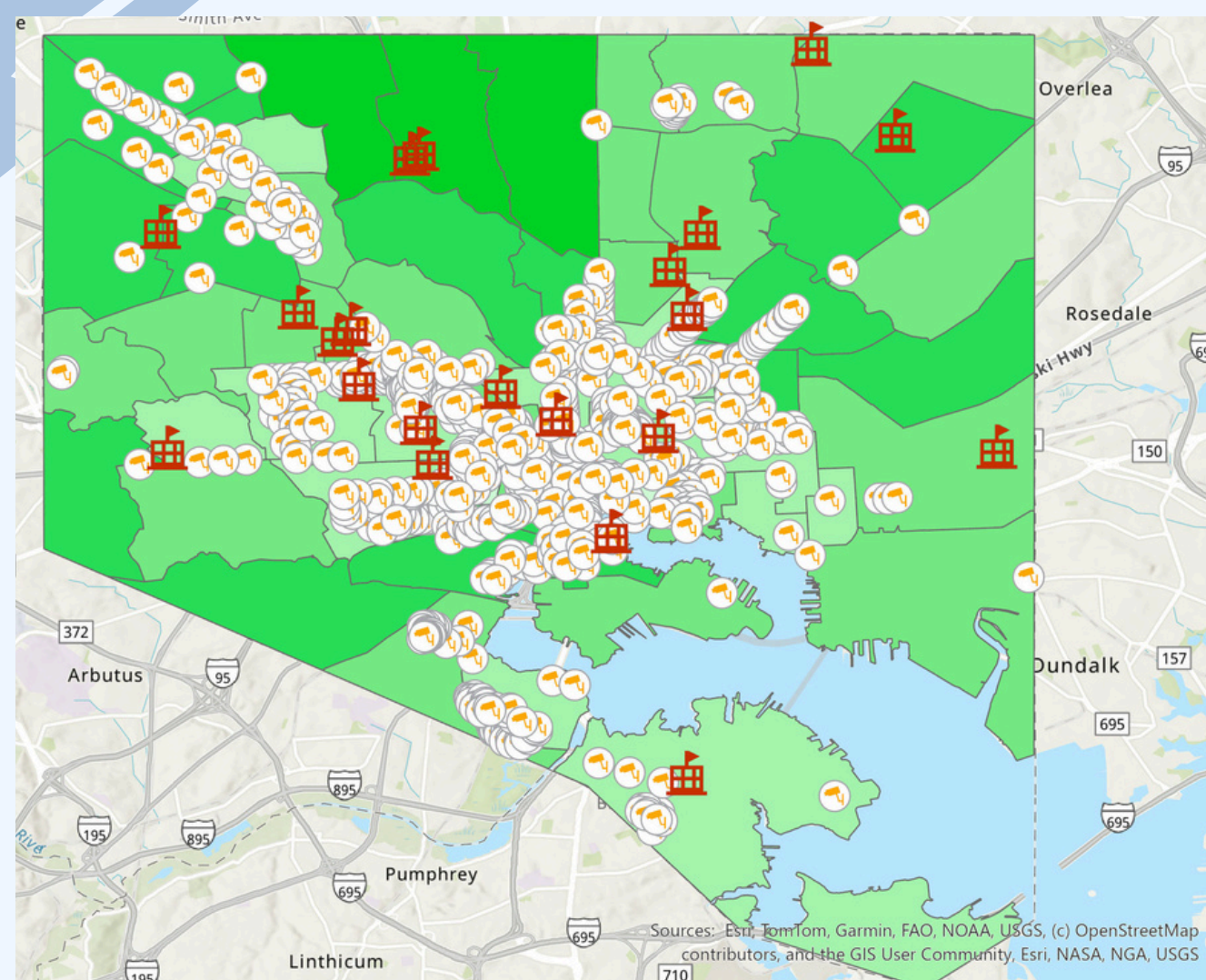
Regression analysis shows no correlation between high school graduation rates and nearby security cameras

Baltimore City: $R^2 = .135$
Baltimore Co. = $1.37E - 06$
DC = .222



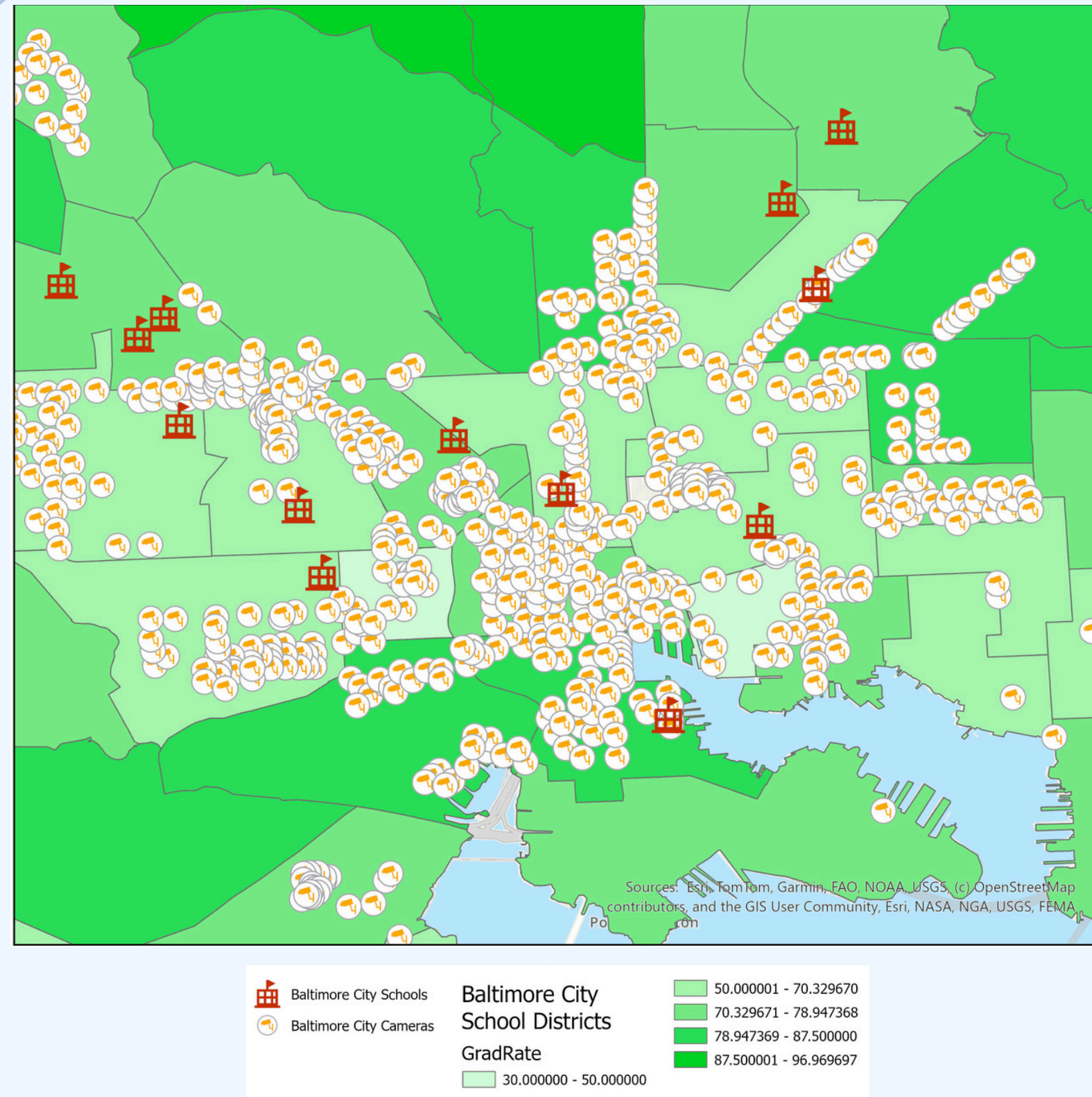
Removing the outliers actually lowers the correlation for DC
More in line with Baltimore City

Results

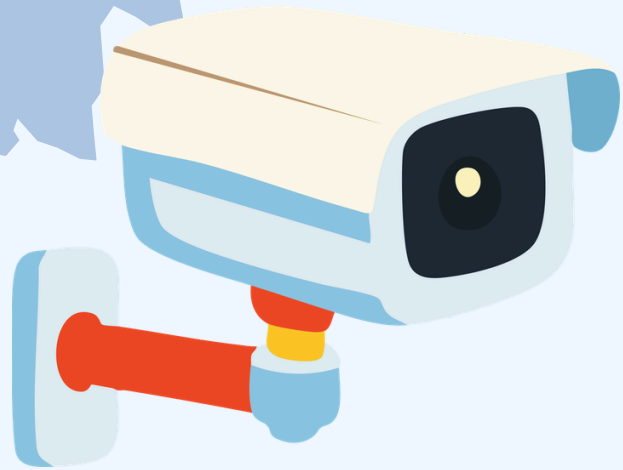


Schools are not doing worse due to proximity or density
Results are more in line with demographic features
Camera proximity does not appear to have an impact

Results



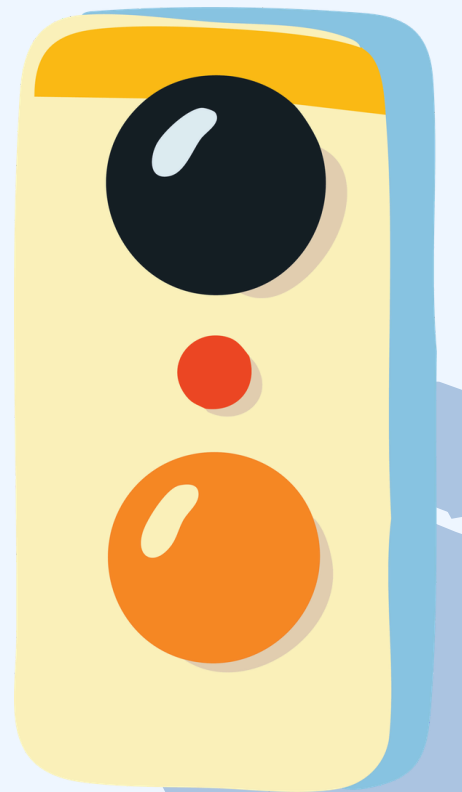
Sheer density of security cameras in Baltimore
Varying proximity, no impact on graduation rates

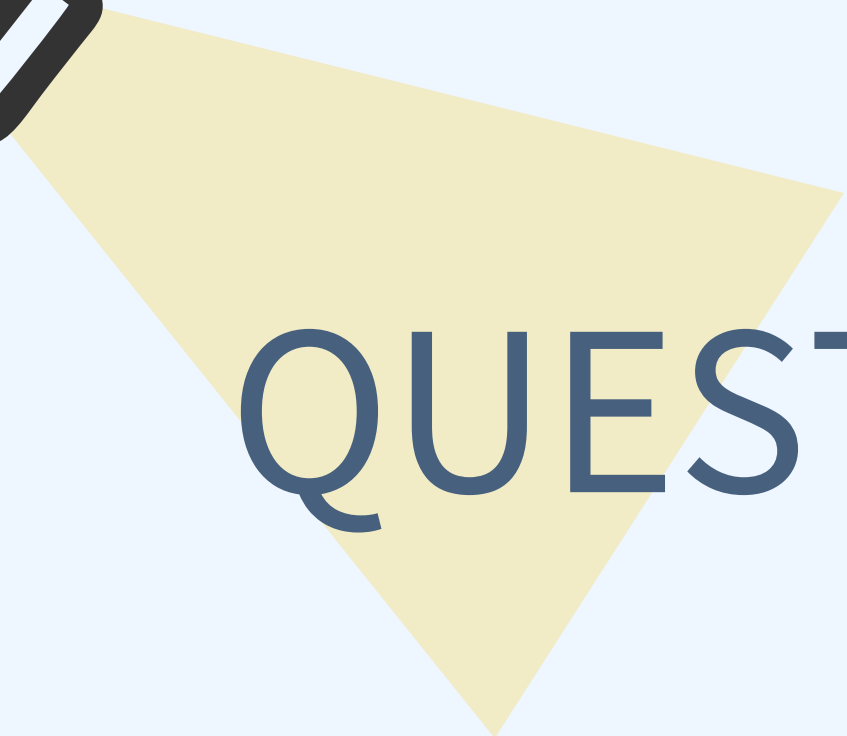
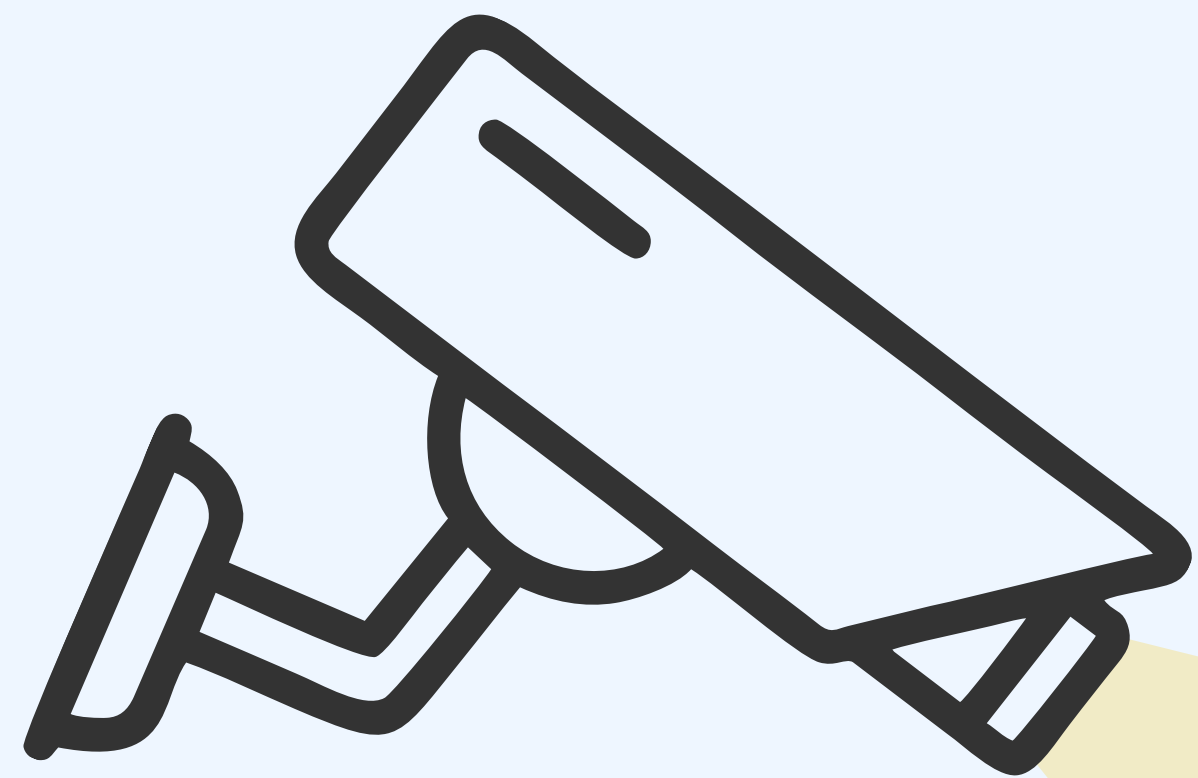


Conclusion

Two possible outcomes from this investigation:

1. There does not exist enough of a 'surveillance culture' outside of schools to affect high school graduation rates
2. The prevalence of security cameras outside of schools does not affect high school graduation rates in the same way security cameras do within schools





QUESTIONS?



References:

- Baltimore City (2025) *Baltimore City Schools* <https://data.baltimorecity.gov/datasets/baltimore::baltimore-city-schools/about>
- Baltimore City (2026) *CitiWatch Camera Locations* <https://data.baltimorecity.gov/datasets/baltimore::citiwatch-camera-locations/about>
- Baltimore County Government (2025) *High Schools* <https://opendata.baltimorecountymd.gov/datasets/BC-GIS::high-schools-1/about>
- Baltimore County Government (2025) *High School Districts* <https://opendata.baltimorecountymd.gov/datasets/BC-GIS::high-school-districts/about>
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- Baltimore Neighborhood Indicators Alliance (2023) *High School Completion Rate - Community Statistical Area* <https://data.baltimorecity.gov/datasets/bniajfi::high-school-completion-rate-1/about?layer=0>
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- Maryland State Department of Education (2023) *Report Card* <https://reportcard.msde.maryland.gov/SchoolsList/Index?l=03>
- Open Data DC (2025) *DC Public Schools* <https://opendata.dc.gov/datasets/DCGIS::dc-public-schools/about>
- Open Data DC (2023) *School Attendance Zones (Senior High)* <https://opendata.dc.gov/datasets/DCGIS::school-attendance-zones-senior-high/about>
- Open Street Maps *Surveillance Under Surveillance* <https://mapcomplete.org/surveillance.html>